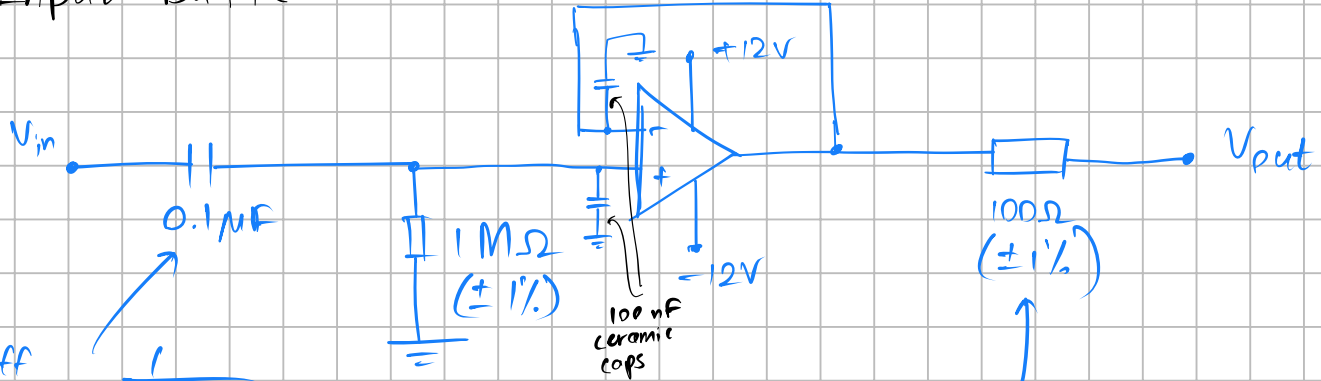


01) Input Buffer



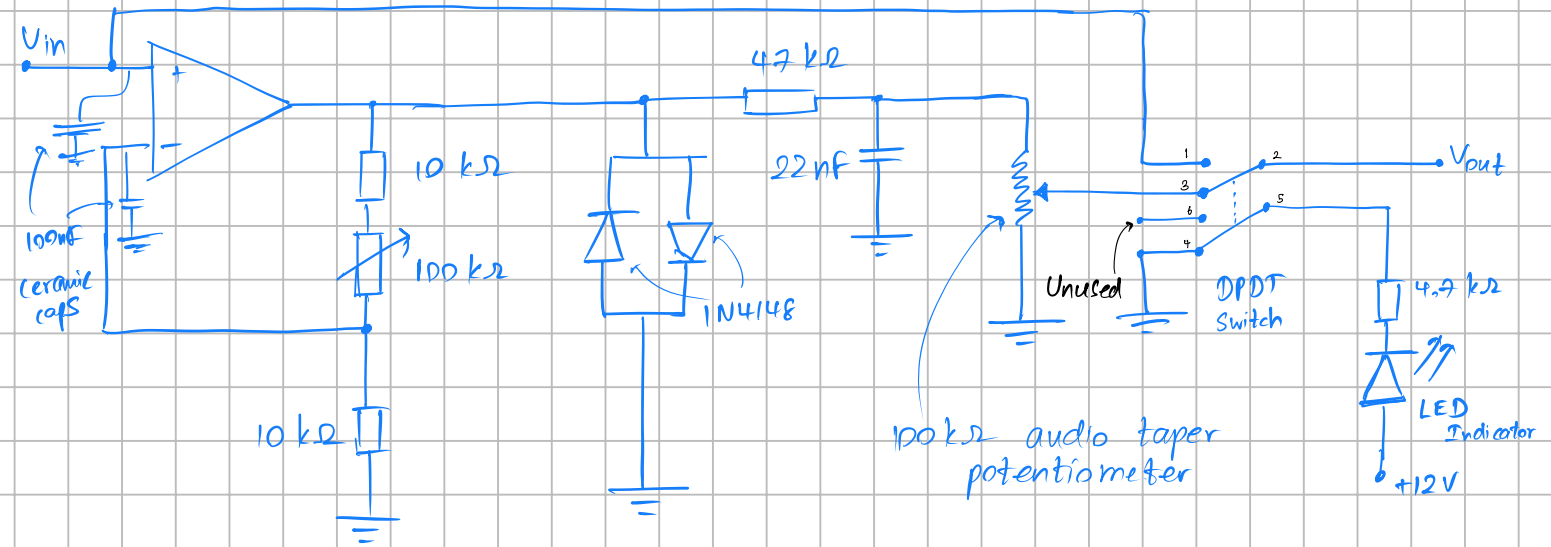
Cut off frequency = $\frac{1}{2\pi RC}$
 $= 1.6 \text{ Hz}$

$R_{in} \approx 1 \text{ M}\Omega$
 $(\because \text{impedance of opAmp})$
 is way too high

To isolate any cable capacitance

- Use schottky diodes in series with power rails for reverse polarity protection. - 1N5819 (40V)
 $\uparrow$
 Breakdown

02) Overdrive / Distortion module



1	0	0	4
2	0	0	5
3	0	0	6

DPDT switch

1 - Module input (From the output of input buffer)

2 (common) - To next module

3 - Output of overdrive module

4 - GND

5 (common) - LED Cathode

6 - Unused

$$\text{Slew rate (Typical)} = 13 \text{ V}/\mu\text{s}$$

$$\text{Maximum slew rate in the circuit} = |2\pi f V_{pk} \cos(2\pi ft)|_{\max}$$

$$= 2\pi \times 20000 \times 1 \times 10^{-6} \text{ V}/\mu\text{s}$$

$$= 0.13 \text{ V}/\mu\text{s}$$

 Component Positions (Step by Step)

 1 Input Connection

On the left, a wire from the buffer output feeds this module.

Before entering the gain stage, you can place a small series resistor (~1 kΩ) to isolate any cable capacitance (optional).

 2 Op-Amp Gain Stage (Hard Clipping Driver)

Place one half of a TL072 op-amp in non-inverting configuration:

Non-inverting input (+) receives the input signal directly.

Inverting input (–) has:

Resistor Rg (10 kΩ) from (–) to ground.

Feedback resistor Rf (220 kΩ) from op-amp output back to (–).

Drive potentiometer: You can replace Rf with a 100 kΩ pot in series with 120 kΩ fixed resistor, so you can vary gain. Physically, draw the pot as a variable resistor symbol between output and (–).

Op-amp power pins go to ±12 V rails (top pin to +12 V, bottom to –12 V) with 0.1 μF decoupling capacitors to ground near the op-amp.

 3 Hard Clipping Diodes

From the op-amp output node, draw two 1N4148 diodes in opposite directions to ground:

One diode’s anode goes to output and cathode to ground.

The other diode’s cathode goes to output and anode to ground.

Position them vertically under the output trace, forming a “back-to-back” pair.

These will clip the waveform symmetrically at $\approx \pm 0.7\text{ V}$ → hard distortion.

 4 Tone Filter

After the diode junction (still on the output line), insert a passive low-pass RC network:

Series resistor (47 kΩ) in the output line.

After that resistor, branch a capacitor (22 nF) down to ground.

This forms a single-pole low-pass to tame harsh highs.

 5 Output Level Control

Place a 10 kΩ or 100 kΩ audio taper potentiometer after the tone filter:

The wiper (middle terminal) goes to the output.

One outer lug receives the filtered signal.

The other outer lug connects to ground.

This pot sets the final output level before the bypass switch.

 6 Bypass Switch (DPDT Toggle or Footswitch)

Draw a DPDT switch block after the level pot:

Top row: one throw connects the effect output to the chain output, the other throw connects input straight through (for bypass).

Bottom row: routes the LED control (or can route ground for true-bypass).

In effect ON position: buffer → op-amp → diodes → tone → level pot → output.

In bypass: input is directly connected to output, completely skipping the overdrive stage.

 7 Indicator LED

Use the other pole of the DPDT:

Connect +12 V through a current-limiting resistor (e.g., 4.7 kΩ) to the LED anode.

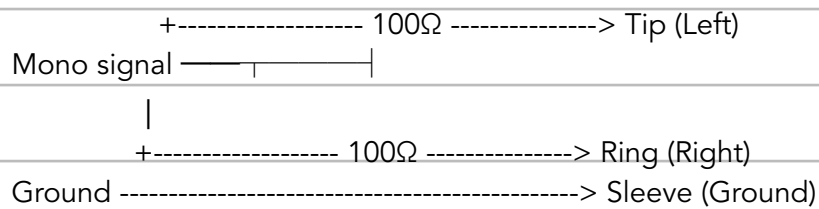
LED cathode goes to the switch, which either connects it to ground (ON) or leaves it floating (OFF).

Position the LED symbol near the switch on the front panel—this gives a clear “Overdrive Active” indication.

 8 Output Connection

After the bypass switch, draw a line to the next module input (e.g., phaser or tremolo).

- Stereo to mono (at output)



- Distortion Module
- Input & Output buffers
- Mono $\leftrightarrow$ Stereo conversion
- PCB Design.